Haozhe Lei

Ph.D. Candidate, Electrical and Computer Engineering | Expected May 2027

RF and Multimodal Sensing | Probabilistic Localization | Robotics and Autonomous Systems

Brooklyn, NY | hl4155@nyu.edu | Homepage | Google Scholar

SUMMARY

ECE Ph.D. candidate developing physics-informed machine learning and probabilistic spatial inference for RF sensing, localization, and autonomous systems. Leads algorithm and neural-model design, Python/PyTorch research prototypes, and evaluation across wireless digital twins and real robotic experiments. Connects uncertainty-aware RF perception with learning-based navigation and adaptive wireless decision-making.

Research record: 13 peer-reviewed journal/conference papers and 5 manuscripts, with first/co-first and corresponding-author leadership across core research.

TECHNICAL SKILLS

Programming and compute: Python, PyTorch, MATLAB, R; Linux, shell scripting, Kubernetes
Probabilistic inference: Likelihood and conditional-posterior estimation, Bayesian multi-observation fusion, Monte Carlo marginalization, uncertainty calibration
Machine learning and decisions: U-Net, attention and Transformer models; physics-informed and neuro-symbolic reinforcement learning, PPO, recurrent actor–critic methods, bandits
Simulation and modeling: Sionna RT, wireless digital twins, real-to-sim calibration, measured antenna-pattern integration, TX/RX pose and mobility simulation, Wireless InSite
Robotics and RF sensing: TurtleBot4/ROS 2 RF–sensor synchronization; Jackal UGV FR3 payload hardware and 3D LiDAR/RGB-D sensing; SLAM-based navigation; Pi-Radio SDR channel sounding; TI radar point-cloud filtering; CARLA, SUMO

EDUCATION

New York University Expected May 2027
Ph.D., Electrical and Computer Engineering
Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS | Ernst Weber Fellowship, 2023

New York University May 2022
M.S., Computer Engineering

China Agricultural University June 2019
B.E., Electrical Engineering and Automation

RESEARCH AND ENGINEERING EXPERIENCE

NYU WIRELESS, New York University *Brooklyn, NY*
Graduate Research Assistant Sept. 2022 – Present

- **Probabilistic RF localization and efficient inference:** Lead model-based likelihood scoring and neural posterior estimation for RF source localization under sparse observations, partial maps, and propagation-model mismatch (*MC-CLE*, *LOCUS-DT*, *MAPLE-RF*). Design U-Net/attention spatial estimators, Bayesian evidence fusion, and uncertainty calibration. MAPLE-RF achieved a **218× fresh-query speedup** over full-grid general-purpose ray tracing in a single-query H100 simulation benchmark, excluding shared RF estimation.
- **Robotic sensing and real-to-sim calibration:** Designed *MAGNETAR*'s joint RF transmitter-position/heading posterior; improved posterior quality over simulation-only training using measurement-aligned simulation and limited real-data adaptation. Coordinate *MobiFR3* architecture and experiments, synchronizing Pi-Radio RF with TurtleBot4/ROS 2, LiDAR, RGB, and odometry. The latest Jackal configuration adds an FR3 payload and 3D LiDAR/RGB-D sensing. Mentor students in RF operation and data collection; validate algorithms on held-out physical measurements.

- **Learning-based control and autonomous navigation:** Lead multi-band digital-twin and learning research (*MCMB-HDT*), designing Transformer rate prediction from sparse, asynchronous histories and recurrent PPO policies for antenna selection under partial observability and sensing costs. Co-developed *PIRL* and symbolic program priors for zero-shot navigation in unseen indoor layouts, using physics-informed rewards and ray-traced propagation cues.
- **Research leadership and spatial intelligence:** Mentor Ph.D. and M.S. researchers across wireless learning, RF–robot experiments, and object-centric RGB-D scene memory. Guide method development, experimental evaluation, and embodied-agent pipelines using vision–language models (VLMs), depth-grounded 3D projection, cross-view association, and scene retrieval.

ADDITIONAL RESEARCH AND INDUSTRY EXPERIENCE

NYU C2SMARTER Center *Brooklyn, NY*
Graduate Researcher Sept. 2023 – Mar. 2026

- Developed distributed online optimization and transportation digital-twin methods for predictive camera control and cooperative driver-risk sensing; contributed to papers in IEEE Control Systems Letters, IEEE T-ITS, and *Transportation Research Part C*.

NYU Center for Cybersecurity *Brooklyn, NY*
Graduate Researcher Sept. 2021 – Jan. 2024

- Developed game-theoretic and neuro-symbolic learning for nonstationary autonomous driving, secure route recommendation, and adaptive penetration testing; contributed to papers at IEEE ICRA, IEEE TIFS, and IEEE MILCOM.

JD.com, Inc. *Beijing, China*
Algorithm Engineer Intern Jan. 2021 – Feb. 2021

- Developed CenterNet/PIoU-based rotated-object detection and supported system integration and on-site validation in a manufacturing laboratory.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Representative work in localization, robotics, and wireless sensing; complete record on Google Scholar.

Selected manuscripts

- **MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band.** First and corresponding author; IEEE ICRA 2027, under review. **[paper]**
- **MAPLE-RF: Efficient Probabilistic RF Source Localization in Partially Explored Environments.** First and corresponding author; IEEE ICRA 2027, under review. **[paper]**
- **Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins.** First and corresponding author; revised and resubmitted to IEEE TWC.
- **Learning a Measurement-to-Posterior Map for Wireless Localization.** First and corresponding author; IEEE TVT, under review.

Selected peer-reviewed publications

- **LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins.** First and corresponding author; IEEE GLOBECOM 2026, accepted. **[paper]**
- **Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning.** Co-first author; IEEE ICRA 2024. **[paper]**
- **Transformer-Based Rate Prediction for Multi-Band Cellular Handsets.** Co-first and corresponding author; IEEE ICC Workshops 2026. **[paper]**

SELECTED HONORS

Ernst Weber Fellowship, NYU Tandon School of Engineering (2023) | School of Engineering Fellowship, NYU Tandon (2022) | Tandon Research Excellence Exhibit (2022)